# AWA Requirements

This is a form that would be filled in by the user to calculate the area of construction required for a project. The user would be filling in only the values that would be specified and then the calculated values would used to calculate the rates of the sections of each phase of the construction. The main requirement is to fill in the values in each of the section, calculate their values, show the totals, then take these values into a separate configuration to calculate the rates for each of them. The rates will be predetermined so it would just be a pick and drop of the values to calculate the final rates. The main form to feed the areas itself contains multiple sections and sub sections under them so we would be required to calculate totals of those sub sections.

Once these are ready, they need to be neatly formatted to present to the client, so two options must be made available- one pdf with a summary, another excel sheet that would contain a main summary sheet just like the pdf along with separate sheets with each of the form and rates values that were input and calculated. It must looks neat, formatted, easy to read, pretty.

## Form Requirements

I will list down fields and sections and sub sections and their requirements. This would be in a format like “field”, “calculation specific”, “measurement in CFT or SqFt or just numbers”. For some of the sub sections I might need to add multiple values in, for example: footings would need multiple sub values of “description”: “area specs” “total”. And this sub section would need a total of all these fields, based on previous example if in footings I have 3 sub values then I need a total of all those values under footings.

1. Centre Line- value in soft
2. Excavation: there will be 3 sub sections in this
    1. Footings: can have multiple sub values of this format- “description”: “area specs” “calculated total”. When user needs to add any, they just click on a + button and it spawns new fields for this.
        1. Area fields needed- Length * Breadth * Depth * Number. This would be calculated in CFt (cubic feet).
        2. Once all fields are filled in a row then show the calculated total of all those fields in the footings sub section.

    2. Compound Wall: only one row
        1. CentreLine * Width * depth = calculated in CFt
    3. Sump: only one row
        1. Length * Breadth * Depth = calculated in CFt
3. PCC 1:4:8: copy same from above Footings leave all values blank. Same calculated values as well.
4. Concrete:
    1. Footings: same fields like footings in excavation. Values will be filled by user. Calculations will be same, provide total of these values in CFt.
    2. Columns: this will also need multiple sub fields like above spec. Values will need same fields as footings. Calculations will be same, provide total of these values in CFt.
    3. Plinth Beam: Central Line * width * depth = calculated in CFt
    4. Lintel concrete: Central Line * width * height = calculated in CFt
    5. Beams concrete: Central Line * width * height = calculated in CFt
    6. Roof Slab: length * breadth * thickness = calculated in CFt
    7. Stairs: 2 calculations needed here
        1. 0.5 * length * breadth * depth * number = calculated in CFt
        2. Length * width * depth = calculated in CFt
    8. Chajja = length * width * number = calculated in SqFt
5. Masonry: Single entry only
    1. Solid Blocks = Central Line * height = Sqft
    2. Bricks = Central line * width * height  = CFt
6. Plastering: 3 sub sections
    1. Inside: Centre Line * height = sqft
    2. Outside: Centre Line * height = sqft
    3. Ceiling: Centre Line * breadth * number = sqft
7. Flooring: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * breadth = calculated in sqft
8. Dadooing: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height = calculated in sqft
9. Doors: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft

10. Windows: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft
11. MS Works: this will have multiple sub sections
    1. TMT Steels: Area * Kilograms = calculated kilograms. User will input both fields.
    2. MS Grill: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft
    3. MS Railing: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft
    4. MS Gate: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft
12. Weather proofing: this will have 2 sub sections, 1 entry each
    1. Terrace = length * breadth = sqft
    2. Bathroom  = length * breadth  = sqft
13. Painting: there will be multiple sub sections in this.
    1. Inside: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft
    2. Outside: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft
    3. Enamel: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft
    4. Polishing: this will need the field specs like footing for description, area fields for multiple rows. There needs to be a calculated total for this for all the row entries of this. The fields needed for these entires will be: length * height * number = calculated in sqft
14. Compound Wall: Length * height  =sqft. Single row entry

15. Sump: value provided by user in Liters
16. Overhead tank: value provided by user in Liters
17. Plumbing works (includes sanitary works, kitchen works): value provided by user in Numbers
18. Fittings: value provided by user in Numbers
19. Electrical works: value provided by user in Area sqft
20. Drainage slab: length * width = sqft. Single entry